Check for updates

Received: 14 December 2024 | Revised: 6 March 2025 | Accepted: 30 March 2025

DOI: 10.1111/aogs.15122

SYSTEMATIC REVIEW

Effects of strength training on quality of life in pregnant women: A systematic review

Paula Redondo-Delgado[1] | Paula Blanco-Giménez[2] | Susana López-Ortiz[1] | Celia García-Chico[1] | Juan Vicente-Mampel[2] | Sergio Maroto-Izquierdo[1]

[1]i+HeALTH, Strategic Research Group, Department of Health Sciences, Miguel de Cervantes European University (UEMC), Valladolid, Spain

[2]Department of Physiotherapy, Medicine and Health Science School, Faculty of Medicine and Health Science, Catholic University of Valencia, Valencia, Spain

Correspondence
Paula Blanco-Giménez, Department of Physiotherapy, Medicine and Health Science School, Faculty of Medicine and Health Science, Catholic University of Valencia, Torrent, Valencia, Spain.
Email: paula.blanco@ucv.es

Funding information
Universidad Católica de Valencia San Vicente Màrtir; Ministry of Education of Spain, Grant/Award Number: 24CO1/001933; Miguel de Cervantes European University, Grant/Award Number: i+HeALTH fundings

Abstract

Introduction: Physical activity is known to improve physical and psychological outcomes in pregnant women. While aerobic exercise is typically emphasized in physical activity guidelines for pregnant women, emerging research suggests that strength training may offer unique benefits beyond those provided by aerobic exercise alone. This systematic review aimed to systematically explore the effects and characteristics of strength training interventions on the health-related quality of life of pregnant women, with the goal of informing more comprehensive and specific exercise guidelines.

Material and Methods: A systematic literature search was conducted across PubMed, Web of Science, and EBSCO Host databases without time restrictions, following PRISMA guidelines (PROSPERO ID: CRD42024511477). Nine randomized controlled trials met the inclusion criteria, involving a total of 1581 participants.

Results: The studies reviewed demonstrated that strength training during pregnancy can mitigate excessive weight gain, alleviate low back and sciatic pain, enhance mood, and improve various aspects of health-related quality of life. These aspects include physical activity levels, muscular strength, flexibility, sleep quality, energy expenditure, and psychological well-being.

Conclusions: Despite these promising findings, this review highlights the need for standardized methodologies and detailed reporting in future research. Incorporating strength training into general exercise recommendations for pregnant women has the potential to optimize maternal health outcomes such as muscle strength, weight gain, physical activity levels, low back pain, pelvic pain, fatigue, anxiety, energy levels, vitality, sleep duration, and health status.

KEYWORDS
aerobic exercise, exercise guidelines, physical activity, pregnant women, quality of life, resistance training

Abbreviations: AE, aerobic exercise; HRQoL, health-related quality of life; PA, physical activity; PW, pregnant women; QoL, quality of life; RCT, randomized controlled trial; ST, strength training; TESTEX, Tool for the assEssment of Study qualiTy and reporting in EXercise.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.
© 2025 The Author(s). *Acta Obstetricia et Gynecologica Scandinavica* published by John Wiley & Sons Ltd on behalf of Nordic Federation of Societies of Obstetrics and Gynecology (NFOG).

1 | INTRODUCTION

It is widely known that regular physical activity (PA) during pregnancy has a positive impact on both physical and mental health of pregnant women (PW). Indeed, the World Health Organization[1] recommends that PW without contraindications should engage in at least 150 minutes of moderate intensity aerobic exercise (AE) per week in accordance with the guidelines from the American College of Obstetricians and Gynaecologists[2] and the Canadian Society of Obstetrics and Gynaecologists.[3] These guidelines also recommend the practice of resistance training during pregnancy.[1–3]

However, PW frequently exhibit sedentary behaviors and reduced PA levels as it navigates the substantial physiological and psychological changes inherent to pregnancy.[4] A sedentary lifestyle in PW can have metabolic consequences or increase the risk of gestational hypertension.[5] Moreover, PW often suffer from a variety of pregnancy-related symptoms and problems, including low back pain,[6] anxiety,[7] depression,[8] or weight gain,[9] some of which may represent a barrier to PA in PW.[10]

Likewise, during pregnancy, several physical, psychological, and social factors have been considered to negatively impact women's quality of life (QoL),[11] which represents a predictor of a country's public health development[12] related to disability in PW.[13]

Specifically, health-related QoL (HRQoL) refers to the subjective perception of physical, mental, and social well-being.[14,15] HRQoL in PW might be influenced by the natural physical and physiological changes associated with pregnancy, where physical performance, mobility, PA limitations, self-care issues, fatigue, and discomfort appear.[14,16,17] Therefore, this outcome is improved in those that have a good mental condition, a reduced pain perception, adequate resting levels and that do physical exercise, have a good mental condition a reduced pain perception, and adequate resting levels.[11]

Recent research highlights that PA during pregnancy has a positive impact on PW health. A previous meta-analysis showed that PW involved in a PA program exhibited a reduced likelihood of experiencing excessive gestational weight gain.[18] Likewise, a comprehensive systematic review highlighted the positive impact of exercise in mitigating the severity of low back pain and lumbopelvic pain.[19] Additionally, evidence suggests that PW who maintain an active lifestyle reported a better health status than their sedentary PW counterparts.[20] Consequently, a growing body of literature supports the promotion of exercise among PW to significantly enhance their HRQoL,[21–23] as this outcome may be related with the improvement of physical outcomes such as increased muscle strength, flexibility, and aerobic capacity. Improved muscle strength helps alleviate physical discomforts and supports functional mobility, both crucial for daily activities during pregnancy.[24] Enhanced flexibility may also reduce the risk of musculoskeletal pain, while aerobic capacity contributes to better energy levels and overall resilience.[25] The increase of muscle strength, flexibility, and aerobic capacity collectively support HRQoL by enabling PW to manage pregnancy-related physical demands more effectively.

Key message

Strength training during pregnancy offers unique benefits, including reduced excessive weight gain, alleviated pain, improved mood, and enhanced health-related quality of life. Incorporating strength training into exercise guidelines for pregnant women could optimize maternal health, but future research needs standardized methodologies and detailed reporting.

The effects of ST programs during pregnancy and the characteristics of ST interventions in PW should be analyzed to develop comprehensive exercise guidelines. There is a need to ensure that PW receive the most effective and safe exercise recommendations tailored to their specific needs and conditions. Although the effects of exercise on HRQoL in PW have been analyzed previously,[23] no study has specifically focused on evaluating the effects of ST. Therefore, this systematic review aimed to assess the effects and characteristics of ST interventions on HRQoL and HRQoL-associated factors (muscle strength, weight gain, PA levels, low back pain, pelvic pain, fatigue, anxiety, energy levels, vitality, sleep duration, and health status)[26] in PWQoL, to enable the creation of specific and precise guidelines.

2 | MATERIAL AND METHODS

2.1 | Search strategy

A systematic review of the literature was performed using the guidelines in the Cochrane Handbook for Systematic Reviews of Interventions (version 6.0)[27] and the checklist for the Preferred Reporting Items for Systematic reviews and Meta-Analyses 2020 (PRISMA).[28] The review protocol has been registered (PROSPERO ID: CRD42024511477).

A systematic computerized search of the literature in PubMed, Web of Science (including Web of Science and MEDLINE results), and EBSCO Host (CINAHL with Full Text, eBook Collection [EBSCOhost], ERIC, Food Science Source, Library, Information Science & Technology Abstracts, MEDLINE Complete, PSICODOC, SPORTDiscus with Full Text). The search timeframe was not restricted (from inception of indexing until December 10, 2024). We developed our search strategy based on the lack of reviews and meta-analyses of the benefits of ST on HRQoL in PW. The search language was restricted to English. A more specific search included the terms of "training," "exercise," "resistance," "strength," "weight," "pregnant," "pregnant women," and "pregnancy". Thus, the following search string was used: ("pregna*"[All Fields]) AND ("resistance training"[All Fields] OR "strength training"[All Fields] OR "weight training"[All Fields] OR "strength exercise"[All Fields] OR "resistance exercise"[All Fields]).

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The reference lists of all selected publications were verified to retrieve relevant publications that were not identified by the computerized search. References to the selected and included original articles, abstracts, and conference proceedings were also searched, including publications, posters, abstracts, and conference proceedings. The search scope was also expanded to clinical trial registry databases (https://clinicaltrials.gov and https://www.clinicaltrialsregister.eu/ctr-search/search).

To identify relevant articles, titles, and abstracts of all selected publications after the first search were analyzed, looking for benefits on quality of life, which included any type of resistance training for PW. In the specific search, in addition to the identified citations of the first search, titles and abstracts of all recognized publications were examined in detail. Full-text papers were recovered if the abstract provided insufficient information to establish eligibility or if the article abstract had passed the first eligibility review.

2.2 | Eligibility criteria

We included original research articles, both randomized controlled trials (RCT) published in peer-reviewed journals that examined the effects of strength exercise interventions on the QoL of PW and were written in English. As reported by the participant, intervention, comparison, outcome, and context (PICOC) process, we used the following criteria to select the studies: (i) participants: healthy PW without any restriction on age; (ii) intervention: long-term (≥6 weeks) strength or concurrent exercise with a minimum training frequency of 2 days per week; (iii) comparison group: non-intervention control group or different form of exercise as control condition; (iv) outcome measures: any component previously established to measure QoL during pregnancy; (v) context: supervised interventions. Only RCTs or secondary analysis that included different outcomes from the main RCT were selected to ensure that the evaluated evidence was based on studies with the highest possible methodological rigor, aiming to provide a clear understanding of the interventions and their effects.

2.3 | Data extraction and summary

All criteria were independently applied by two reviewers (PR-D and SM-I) to the full texts of articles that passed the eligibility screening of titles and abstracts. One researcher (PR-D) double-checked the included studies in the systematic search and modified the eligibility criteria to limit the scope of the review. Disagreements were resolved by discussion. Duplicate references were removed using an online deduplication tool for systematic reviews (https://sr-accelerator.com/#/libraries/dedupe) and subsequent manual methods. Two authors (PRD and SMI) independently screened titles and abstracts to determine initial eligibility using systematic review software (https://www.rayyan.ai). Blinding of the authors was used to reduce bias during this process. Finally, the full text was reviewed to determine eligibility for inclusion based on the eligibility criteria. Disagreements in eligibility decisions were resolved through discussion or by a third reviewer (AS-L), when required.

Data extraction was performed independently and in duplicate by two authors (PR-D and SM-I). The data were then merged by one author (PR-D), and any discrepancies in the extracted data were resolved through discussion or by a third reviewer (SL-O) if required. The extracted data from each full-text article included the following: (i) study identification information; (ii) study design; (iii) sample size; (iv) age, height, and body mass; (v) exercise program characteristics (e.g., program duration, weekly training frequency and training volume, exercises prescribed, exercise intensity, training load management, and supervision); (vi) measures of any component of the quality of life before and after the intervention program; (vii) means and standard deviations (SD) for relevant outcome measures (i.e., pre- and post-test); and (viii) exact *p*-values, *r*-values, *t*-values, or confidence intervals for an association between two outcomes or a comparison between groups. When insufficient data were reported, authors were contacted via email. When data were not presented in tables or text and when authors did not provide the requested data, they were extracted from figures using WebPlotDigitizer (Web Plot Digitizer, V.4.1. Texas, USA) when possible.

2.4 | Methodological quality and risk of bias assessment

Methodological quality and risk of bias were independently assessed by two researchers (PR-D and SM-I) using Cochrane risk of bias 2 (RoB2).[29] In cases of disagreement between the scores, a third author made the final decision (SL-O). The RoB2 assessment scale was structured into a fixed set of domains of bias, focusing on different aspects of trial design, conduct, and reporting. Five domains were assessed: (D1) bias arising from the randomization process; (D2) bias due to deviations from intended interventions; (D3) bias due to missing outcome data; (D4) bias in the measurement of the outcome; and (D5) bias in the selection of the reported results. These categories were classified as having a "high risk of bias," "low risk of bias" or "some concerns." Risk-of-bias figures were created using Robvis software. The risk of bias analysis was complemented by an assessment of methodological quality, as recommended in the PRISMA guidelines.[28] The Tool for the assEssment of Study qualiTy and reporting in EXercise (TESTEX) scale was used for this purpose.[30] This tool was developed specifically for the assessment of the methodological quality and reporting of exercise training studies.[30] The TESTEX scale includes 12 criteria, with a maximum score of 15 points, divided into two categories: study quality (5 points) and reporting quality (10 points). It assesses the following criteria: (1) clear and fulfilled eligibility criteria; (2) randomization methods described and defined; (3) concealment of participant assignment; (4) groups with no statistical differences at pretest; (5) blinding of the evaluator; (6A) more than 85% of participants completing the study; (6B) adverse events reported for each group; (6C) attendance

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

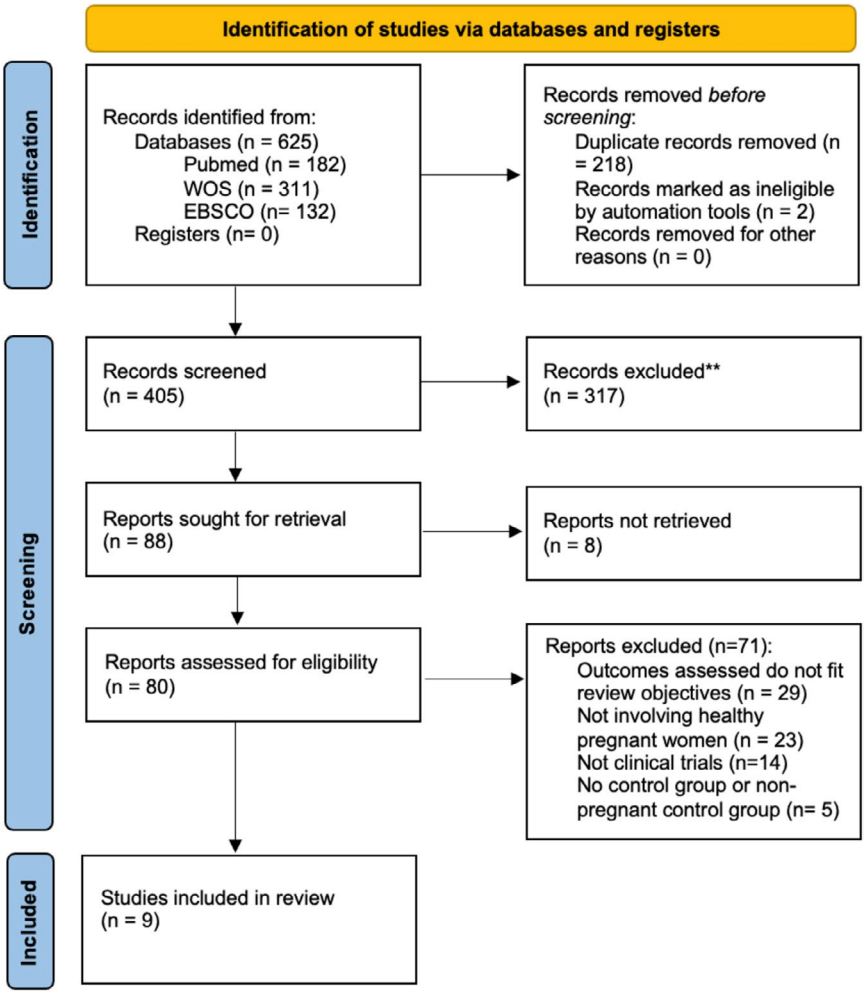

FIGURE 1 PRISMA flow chart for study selection.

at completed sessions reported for participants who completed the study; (7) intention-to-treat analysis; (8A) between-group statistical analysis reported for the primary dependent variable; (8B) between-group statistical analysis reported for the secondary variable(s); (9) outcome variability results reported; (10) PA level of the control group reported; (11) exercise intensity adjusted during the intervention; and (12) volume and energy expenditure calculable.

3 | RESULTS

3.1 | Study selection

The process of study selection according to PRISMA is shown in Figure 1. The initial search of the electronic databases identified 625 titles, of which 218 were rejected as duplicates and two as ineligible. Thus, 405 titles were identified, but 317 were rejected after reading because they did not meet the inclusion criteria. Of the 88 abstracts screened, eight were not retrieved and 71 were excluded because they did not meet the inclusion criteria. Finally, nine studies satisfied the inclusion criteria for this review.[21,31–38] The complete list of excluded references and the rationale for their exclusion can be found in Appendix S1.

3.2 | Characteristics of the included studies

The main characteristics of the studies included in the review, including the participants, interventions, and results, are summarized in Table 1. Of the included RCTs, four were conducted in Spain,[31,32,37,38] three in Norway,[34–36] one in the United States,[21] and one in Sweden.[33]

3.2.1 | Characteristics of the participants

A total of 1581 PW were included, considering that a subset of three studies[34–36] analyzed the same participants but focused on different outcomes. Additionally, one trial[38] accounted for 44% of the total sample. Demographic data from the participants were provided for all studies. The mean age of the participants was 31.64 ± 4.38 years, with a mean age of 31.47 ± 4.26 years on the intervention groups and 31.8 ± 4.5 years for the standard care group. Prepregnancy body mass index information was available in all the studies[21,31–38] with a mean body mass index for all participants of 23.84 ± 4.24 kg m^{-2} and with a healthy weight following the IOM recommendations in both the intervention (23.9 ± 4.43 kg m^{-2}) and control (23.8 ± 4.05 kg m^{-2}) groups. Gestational weight was

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Characteristics of included studies investigating the benefits of strength training on HRQoL-related outcomes and their main findings.

Author/ Year/ Country	Sample and study characteristics	Clinical characteristics	FITT Principles (Frequency, Intensity, Time, Type)	Outcomes and tools	Main findings
Acosta-Manzano et al., 2022; Spain[31]	$n = 121$ healthy PW Duration: From 16–17 to 33–34 week of gestation Data collection: • Pre-intervention (wk 16–17) • Post-intervention (wk 33–34) • 6 wk postpartum	Age (years) • SEP group: 33 ± 5 • CG: 34 ± 5 BMI prepregnancy (kg/m^2) • SEP group: 23.4 ± 6.3 • CG: 22.1 ± 4.4 Weight gain (kg) • SEP group: 9.2 ± 4.4 • CG: 12.6 ± 4.9	SEP group ($n = 67$): • F: 2 days/week ST and 1 day per week AE. • I: moderate to vigorous (RPE 12–16). • T: ST: 3 sets, no information about repetitions. • T: Combined training (inter-session). CG ($n = 54$): Usual care and recommendations to promote healthy pregnancy	Sedentary time and physical activity (CT3X + triaxial accelerometers) Sleep behaviors (PSQI) Physical fitness (upper body muscle strength and CRF) Weight gain and weight retention (digital scale)	SEP group: ↓↓ late weight gain ↓ weight retention = prevent excessive weight gain ↑↑↑ CRF ↑↑ upper body strength ↑ sleep duration ↓ mean weight gain ↓ excessive weight gain values
Aparicio et al., 2022; Spain[32]	$n = 93$ healthy PW Duration: From wk 16 to wk 34 of gestation Data collection: • Pre-intervention (wk 16) • Post-intervention (wk 34)	Age (years) • SEP group: 33.3 ± 4.2 • CG: 33 ± 4.8 BMI prepregnancy (kg/m^2) • SEP group: 24.2 ± 4.1 • CG: 23 ± 3.4 Weight gain (kg) • SEP group: 9.6 ± 4.4 • CG: 12.6 ± 5.1	SEP group ($n = 49$): • F: 3 days per week. • I: moderate to vigorous. • T: 2 circuits of 15 exercises (40 s work/20 s rest). • T: Concurrent training (intra-session): ST alternating with short bouts of AE CG ($n = 44$): Usual activities and a series of lectures to address the importance of PA and a healthy diet during pregnancy	Low back and sciatic pain (VAS) Pain disability (ODI)	SEP group: ↓↓↓ VAS low back pain ↓↓ VAS sciatica score ↓↓ pain ODI total score CG: ↑↑ pain while sleeping, ↑↑ pain while lifting weight ↑↑ limitations of the social life due to pain
Fieril et al., 2015; Sweden[33]	$n = 72$ healthy PW Duration: From wk 14 to wk 25 of gestation Data collection: • Pre-intervention (wk 13) • Post-intervention (wk 25)	Age (years) • SEP group: 30.8 ± 3.6 • CG: 30.6 ± 3.4 BMI prepregnancy (kg/m^2) • SEP group: 23 ± 2.6 • CG: 22.6 ± 2.5	SEP group ($n = 38$): • F: 2 days per week • I: moderate to vigorous • T: 50–80 repetitions for each muscle group, during 3–5 min, including short rests • T: Concurrent training (intra-session): ST and AE (inspired by BODYPUMP™) CG ($n = 34$): usual activities, usual care and recommendations to promote healthy pregnancy	Quality of Life (SF-36) Pain (graphic depiction by body locations) Gestational body weight gain (digital beam scale) Handgrip isometric strength (electronic handgrip strength detector) Functional status (DRI) Physical activity level (IPAQ)	= HRQoL between groups, neither number of women with back pain nor those with pelvic pain = group-means scores for peak voluntary strength, weight gain and activity level ↓ functional status deteriorated during the intervention in both groups and ↑ pain

(Continues)

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 (Continued)

Author/ Year/ Country	Sample and study characteristics	Clinical characteristics	FITT Principles (Frequency, Intensity, Time, Type)	Outcomes and tools	Main findings
ClinicalTrials.gov Protocol Registration System NCT00617149					
Haakstad et al., 2011; Norway[35]	$n = 105$ healthy PW Duration: From 12 to 24 wk of gestation to 36–38 wk of gestation	Age (years) • SEP group: 31.2 ± 3.7 • CG: 30.3 ± 4.4 BMI prepregnancy (kg/m^2) • SEP group: 23.8 ± 3.8 • CG: 23.9 ± 4.7 Weight gain (kg) • SEP group: 13.8 ± 4 • CG: 12.5 ± 4	SEP group ($n = 52$): • F: 2 days per week • I: Moderate to vigorous (RPE: 12–14 somewhat hard) • T: No information • T: Concurrent training (intra-session): AE (dancing with low-impact exercises and step training) and ST (deep abdominal stabilization muscles, pelvic floor, and back muscles) CG ($n = 53$): Usual PA pattern	Gestational body weight gain (digital beam scale and skinfold thickness)	SEP group (all sessions): ↓↓ weight gain ↓ weight retention 6–8 wk postpartum = between groups skinfold thickness
Haakstad et al., 2015; Norway[36] Secondary analysis	$n = 105$ attendance to ≥80% sessions (intervention: 52; control: 53)			Pelvic girdle pain (interview questionnaire) Low back pain (interview questionnaire)	= between groups in pelvic girdle and low back pain prevalence at any assessment points were showed. No negative effects of the 12 wk intervention program were reported
Haakstad et al., 2016; Norway[34] Secondary analysis	$n = 105$ attendance to 24 sessions (intervention: 52; control: 53) Data collection: • Pre-intervention (wk 12–24) • Post-intervention (wk 36–38) • 6–12 wk postpartum			Psychological outcomes (16 questions adapted from two questionnaires: WHOQOL-brief and SF-36) Common pregnancy complaints (interview questionnaire)	SEP group: ↑↑ health status ↓ fatigue related to daily activities SEP group (all sessions): ↓↓ sadness, hopelessness or anxiety ↑ energy levels CG: ↑↑ life enjoyment = between groups in body-image, pregnancy depression nor pregnancy complaints
Marín-Jiménez et al., 2023; Spain[37]	$n = 94$ healthy PW Duration: From wk 16 to wk 34 of gestation Data collection: • Pre-intervention (wk 16) • Post-intervention (wk 34)	Age (years) • SEP group: 33 ± 4.3 • CG: 33.3 ± 4.9 Weight gain (kg) • SEP group: 9.66 ± 4.6 • CG: 12.5 ± 5.3	SEP group ($n = 45$): • F: 2 days per week ST and 1 day per week AE • I: Moderate to vigorous (RPE: 12–16 and HRR 65%) • T: ST: 3 sets, no information about repetitions • T: Combined training (inter-session): ST (hip and knee dominant exercises, pull and push movements and core muscle exercise) and AE (short bouts) CG ($n = 49$): Usual activities and a series of educational lectures	Gestational weigh gain Health-related Quality of Life (SF-36)	SEP group: ↓↓ SF-36 physical functioning domain, ↓↓ SF-36 physical component summary and ↓↓ SF-36 social functioning

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 (Continued)

Author/ Year/ Country	Sample and study characteristics	Clinical characteristics	FITT Principles (Frequency, Intensity, Time, Type)	Outcomes and tools	Main findings
O'Connor et al., 2018; United States[21]	$n=134$ healthy PW Duration: From 19 to 24 wk of gestation to 36–38 wk of gestation Data collection: • Pre-intervention (wk 19–24) • Post-intervention (12 wk post-intervention)	Age (years) • SEP group: 28 ± 5 • CG: education: 29 ± 5 and wait list: 29 ± 5 BMI prepregnancy (kg/m^2) • SEP group: 26 ± 6 • CG: education: 27 ± 5 and wait list: 28 ± 5	SEP group ($n=44$): • F: 2 days per week. • I: Moderate to vigorous (RPE) • T: 2 sets of 15 reps of 6 resistance exercises, low velocity (~2 s concentric and ~2 s eccentric) and with 1 min rest period between sets and 2 min between exercises • T: ST (leg extension, leg press, arm lat pull, leg curl, lumbar extensions, and an abdominal exercise) CG ($n=90$): • Education ($n=45$): a series of educational classes • Wait list control ($n=45$): no treatment control group	Quality of Life (SF-36) Mood (POMS)	SEP group: ↑ vitality. ↓ fatigue. Adverse changes in symptoms of energy and fatigue during pregnancy were ↓ by adopting low-to-moderate intensity ST
Ruiz et al., 2013; Spain[38]	$n=962$ healthy PW Duration: From 5 to 6 wk of gestation to 38–39 wk of gestation Data collection: • Pre-intervention (wk 5–6) • Post-intervention (before delivery)	Age (years) • SEP group: 31.6 ± 4 • CG: 31.9 ± 4 BMI prepregnancy (kg/m^2) • SEP group: 23.7 ± 3.9 • CG: 23.5 ± 4.2 Weight gain (kg) • SEP group: 12.3 ± 3.6 • CG: 13.8 ± 3.8	SEP group ($n=481$): • F: 2 days per week ST and 1 day per week AE. • I: Moderate to vigorous (RPE 10–12 and HR) • T: ST: no information, AE: periods of 3–4 min. • T: Combined training (inter-session): ST (major muscle groups) and AE (low-impact aerobic dance, involving the upper and lower limbs.) CG ($n=481$): Usual activities, usual care and recommendations to promote healthy pregnancy	Gestational body weight gain (BMI)	SEP group: ↓↓↓ weight gain and were ↓↓ likely to gain weight above the IOM recommendations

Note: Symbols: ↓ indicates a significant change with a p-value greater than 0.05, ↓↓ indicates a larger significant change with a p-value greater than 0.01, and ↓↓↓ indicates an even more substantial significant change with a p-value less than 0.001; = means that no significant changes were observed.

Abbreviations: AE, aerobic exercise; BMI, body mass index; CG, control group; CRF, cardiorespiratory fitness; DRI, disability rating index; HR, heart rate; HRR, heart rate recovery; IPAQ, International Physical Activity Questionnaire; min, minutes; ODI, Oswestry Disability Index; POMS, profile of mood states; PSQI, Pittsburgh Sleep Quality Index; reps, repetitions; RPE, rate of perceived exertion; SEP, supervised exercise program; SF-36, short form-36 health survey; ST, strength training; VAS, visual analog scale; WHOQOL-brief, World Health Organization Quality of Life brief; wk, week.

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

assessed in five of the nine studies.[31,32,35,37,38] Mean gestational weight gain (i.e., weight at the 34th gestational weight – prepregnancy) was 11.85 ± 4.41 kg for all participants, with a mean of 10.9 ± 4.2 kg for the intervention group and 12.8 ± 4.62 kg for the control group.

3.2.2 | Outcomes assessed

All studies measured at least one component related to QoL (sedentary time,[31] PA,[31] sleep behaviors,[31] muscle strength,[31,33] weight gain and weight retention,[31,33,35,37,38] pain,[32,33,36] functional status,[21,33,37] psychological outcomes,[34] and mood[21]). In addition, most studies (6/9) used questionnaires to evaluate the HRQoL.[21,31–34,37] The Short Form Health Survey [SF-36] is the only questionnaire that has been used in several articles.[21,33,34,37] Other questionnaires and indexes were used: the Pittsburg Sleep Quality Index Questionnaire,[31] Oswestry Disability Index,[32] Disability Rating Index,[33] International Physical Activity Questionnaire,[33] WHOQOL-brief[34] and Profile of Mood States.[21]

3.2.3 | Characteristics of the interventions

Exercise interventions lasted 12 weeks or more, with a training frequency ranging from two to three sessions per week. All studies included an ST program, but seven included AE,[31,32,34–38] and not all provided full details about the ST intervention (e.g., not specified exercises,[32–36] volume,[31,34–37] or material[31,32,34–37]). All studies structured the training sessions with a warm-up, followed by a principal strength intervention, and a cool down.[21,31–38] The duration of each session ranged from 50[38] to 60 min.[21,31–37] Training sessions are usually conducted at health facilities,[31,32] antenatal clinics,[33] hospitals,[31] or universities.[21,37]

Training sessions were supervised in all studies, and most were based on the ACOG.[31,32,34–38] The interveners were specialized multidisciplinary teams,[31,32] physiotherapists,[33] certified fitness instructors,[34–37] and sports science specialists.[21,38] Only three studies applied a familiarization to teach fundamental basic movement patterns and theoretical explanations of the prescribed exercises.[31,32,37]

All studies performed moderate intensity AE for 30–40 min[21,31–38] per session with a frequency of 1–2 sessions/week, those sessions included exercises such as dance or choreographies,[31,34–36,38] cardiovascular circuit or AE bouts,[31,32,37] interval walking,[32] proprioceptive and coordinative circuits,[32] and step training.[34–36]

All studies used circuit-based ST prescribed based on sets and repetitions[21,31,37] or time.[32] In ST engaged, the main muscle groups were trained functional strength exercises with elastic bands,[38] barbells,[33,38] weight plates,[33] and guided machines.[21,38] The overall aim of ST in these studies was to promote good posture, prevent back pain, and strengthen muscles used in labor as well as the core and pelvic floor.[21,33,38] Two of the studies[31,37] referenced another study[39] to explain the exercises used, including a hip-dominant exercise (e.g., deadlift, hip hinge, or hip swing), a dominant knee (e.g., squats, lunges), two pulling movements, a pushing movement (adapted push-ups) and a core muscle exercise. Other studies included leg extension, leg press, arm lateral pull, leg curl, lumbar extension, abdominal exercises,[21] biceps curls, arm extensions, arm side lifts, shoulder elevations, bench presses, seated lateral row, lateral leg elevations, leg circles, knee extensions, knee (hamstring) curls, and ankle flexions and extensions.[38] Moreover, not all studies performed some type of core and pelvic floor exercises[21,32–36,38] due to their high importance in this population.[21] The external load was measured in seven studies using the 6–20 Borg scale for rate of perceived exertion (RPE)[21,31,34–38] and had to be somewhere between 12 and 14 (somewhat hard) or 15–16 (hard) for PW.[21,31,34–38] All results of the included studies are synthetized in Figure 2.

The control group received standard care only. They were encouraged to continue their usual activities[31,32,37] and sometimes tried to incorporate PA.[33–36] Participants in the control group had the opportunity to attend a series of lectures to address the importance of PA and a healthy diet during pregnancy.[31–33,37,38] In addition, the lectures covered several topics, including information about what to expect during delivery, parent skills, and child development.[21] In addition, in all the studies, the intervention group had the opportunity to attend the same talk.

3.3 | Methodological quality and risk of bias results

The proportion of studies identifying bias in Figure 1 or more design element(s) is presented in Figure 3 and The RoB2 scores of the included studies are presented in Appendix S2. Concerns about underlying bias were mainly due to issues arising from the randomization process and deviations from intended interventions, such as lack of clarity regarding blinding (e.g., whether participants and investigators were aware of their assigned intervention during the trial). However, the risk of bias was low for missing outcome data, which includes attrition bias, outcome measurement, and reported outcome selection bias in all included articles. In addition, the results on methodological quality are presented in Appendix S3. To summarize, 6 out of 9 studies achieved a score of more than 10/15 (1 study achieved 13/15,[33] 3 studies achieved 12/15,[34–36] and 2 studies achieved 11/15,[21,31]) while 3 out of 9 studies achieved scores between 5 and 10/15 (1 study achieved 9/15[37] and 2 studies achieved 10/15.[32,38]) The criteria most frequently not met by the studies were criteria 4, 6A, 10, and 12.

4 | DISCUSSION

This systematic review aimed to provide an overview of the available literature that investigated the effectiveness of ST in PW. Nine RCTs were analyzed in this review. A significant finding of this study

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

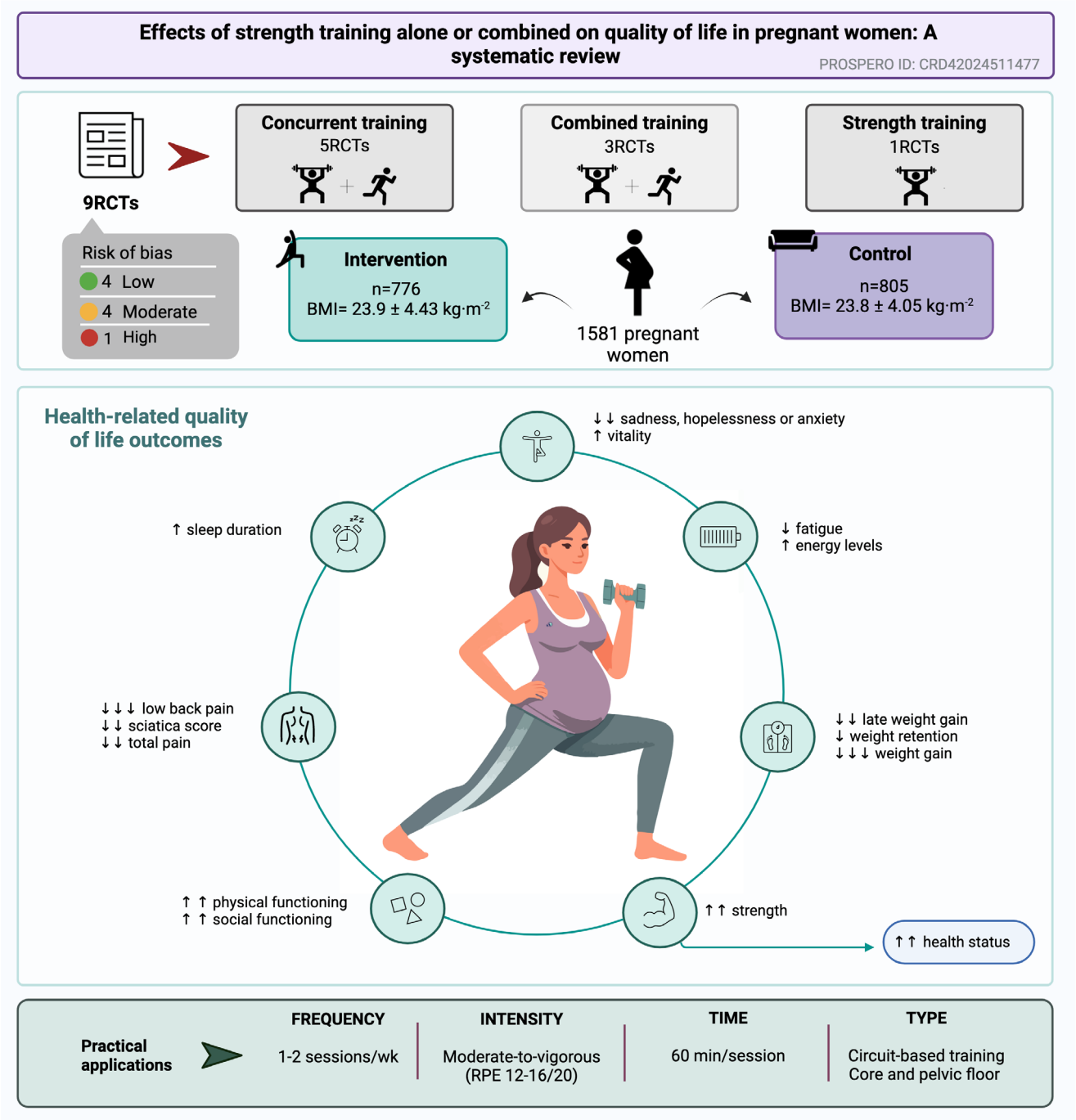

FIGURE 2 Graphical summary of the study findings on the effects of strength training during pregnancy on quality of life-related outcomes.

was the positive impact of ST interventions on enhancing PA levels in PW, encouraging these women to engage in regular PA, which would contribute to improved HRQoL outcomes.[37] Another potential finding was that performing ST 1–2 days per week during pregnancy improved cardiorespiratory fitness, muscle strength, sleep duration, and prevention of excessive weight gain.[31,38] Furthermore, PW who attended all ST sessions demonstrated significant weight gain reduction, increased energy levels[34,35] and consequently, improved HRQoL.

Maternal well-being encompasses the physical, emotional, and mental health of PW, which can be influenced by significant physiological and hormonal changes inherent to pregnancy. Specifically, HRQoL may be enhanced when there is an adequate physical and mental condition,[39] absence of pathological signs,[40] reduced perceptive pain levels,[40] and sufficient resting levels.[41] When executed appropriately and with medical approval, ST can positively impact maternal well-being. It also contributes to maintaining muscle strength,[31] improving posture,[32] reducing low back pain,[32] and

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

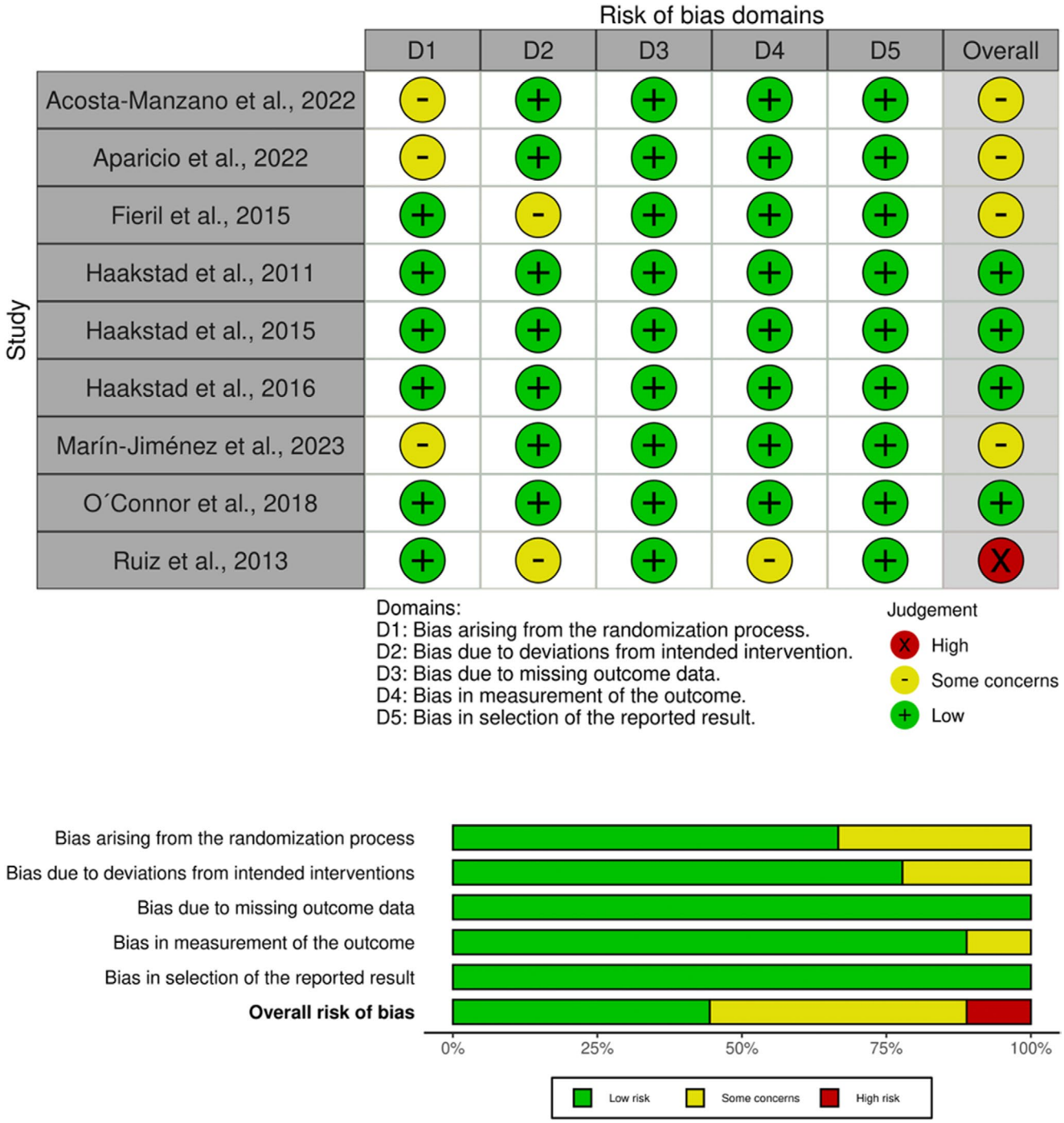

FIGURE 3 Overall risk of bias of articles included in a systematic review of randomized controlled trials investigating effects of strength training in pregnancy women.

increasing energy levels.[21,34] Furthermore, it assists in managing weight gain and enhancing psychological well-being by reducing anxiety and improving mood.

Many PW experience sleep disturbances due to physical discomfort, stress, and psychological factors. Regular PA has been demonstrated to mitigate stress, anxiety, and symptoms of depression and can potentially prevent prenatal and postnatal depression through the release of endorphins,[33] contributing to a more positive pregnancy experience.[42–44] Furthermore, PW who engaged in ST reported significantly higher self-rated health status, with reduced fatigue and improved scores for sadness, hopelessness, and anxiety[21,34] compared to their sedentary counterparts. Additionally, ST can enhance sleep quality by regulating sleep patterns and alleviating insomnia, which is prevalent during pregnancy.[31,32] These beneficial effects contribute to improved emotional stability and an enhanced sense of well-being.[45]

Weight gain is a normal component of pregnancy; however, excessive weight gain and retention can lead to complications such as

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

gestational diabetes, hypertension, and an increased risk of cesarean delivery.[3,46] One of the most prevalent symptoms exacerbated during pregnancy related to weight gain is low back and pelvic pain.[32] As a preventive therapy, ST can help PW to control their weight more effectively,[31] thereby promoting a healthier pregnancy,[31,35] preparing the body for labor, and accelerating postpartum recovery.[47,48] Furthermore, regular ST helps regulate blood glucose levels and improves insulin sensitivity, reducing the risk of gestational diabetes and associated maternal and fetal complications,[49,50] such as preeclampsia.[51,52] Regarding low back and pelvic pain, ST strengthens the muscles supporting the spine and pelvis, which may alleviate this discomfort and improve posture,[32,33] although some studies did not demonstrate significant differences in perceived pain between groups.[33,36] In summary, ST during pregnancy plays a crucial role in managing weight gain, which may positively affect HRQoL. However, these findings should be interpreted with caution, as only four studies analyzed weight gain, and one study[38] accounts for 44% of the total sample analyzed, potentially influencing the generalizability of the results.

Regarding physical function, it is a key indicator of maternal health and is influenced by improvements in neuromuscular factors such as muscle strength and muscle endurance (e.g., sit-to-stand test), commonly achieved through ST.[31,39] These variables are closely associated with the ability to perform activities of daily living, such as carrying weight, maintaining balance, or ascending stairs. Enhancing these functional outcomes facilitates the performance of daily activities for PW and reduces physical strain during pregnancy.[31] Improved physical health and energy levels enable PW to participate more actively in social activities.[53] Monitoring these parameters can identify potential risks, allowing for targeted interventions to support mobility and prevent musculoskeletal complications during pregnancy. However, there is limited information on the specific strength tests and variables previously assessed in existing studies. Handgrip strength is the most frequently mentioned measure, as it correlates with overall upper extremity strength.[31,54] However, only one study included in this systematic review performed neuromuscular strength assessment, specifically the handgrip strength test.[33] Despite the fact that various tests for measuring strength and functional capacity in PW have been proposed and validated throughout the literature, such as the five-repetition sit-to-stand mechanical power test,[55] most studies lack specific strength tests to substantiate the benefits of ST on neuromuscular capacity, even though it is well-correlated with health-related quality of life (HRQoL) in PW. Therefore, further research is needed to incorporate a broader range of neuromuscular and functional assessments, such as load-velocity profile, balance, and gait analyses, during different trimesters of pregnancy to comprehensively monitor maternal health and the effectiveness of ST interventions.

Finally, the results of QoL questionnaires in PW demonstrate the significant impact of physical, emotional, and social changes during this period, which have shown considerable improvement through ST. They increase vitality and energy[21] and reduce fatigue,[21] sadness or anxiety[34] and even pain,[32] which leads to an improvement in social life and functional status.[34,37] Understanding these outcomes allows healthcare providers to address specific needs, improving both PW HRQoL and overall pregnancy outcomes.

Overall, the studies demonstrated moderate- to high methodological quality, with most scoring above 10/15 on the TESTEX scale. However, limitations in terms of risk of bias were noted, particularly in the randomization process and deviations from intended interventions where incomplete allocation concealment and lack of blinding may have influenced subjective outcomes such as self-reported quality of life. Despite these problems, the risk of bias in missing data, outcome measurement, and reported results was generally low. Methodological shortcomings included insufficient information on adverse events, participation in the training sessions, and adjustment of training intensity, which were treated inconsistently across studies. These limitations should be considered when interpreting the results, as they may overestimate the effects of ST on HRQoL. In addition, our study has further limitations. First, data from studies with other study designs different from RCTs were not considered.[53,56] These studies might have provided further data on the significance of the effects of PA interventions. Second, only one RCT included PW at the first trimester of gestation.[38] Therefore, the evidence of ST in this pregnancy phase is still scarce and the results should be considered with caution when applied to those PW. Third, weight gain does not differentiate between fat and muscle, which is important concerning in ST. Fourth, the outcomes assessed in the studies included are diverse, which made it difficult to meta-analyze the results. This gap in the literature restricts the ability to provide comprehensive recommendations for PA interventions that could optimize health outcomes for both the mother and the fetus. Fifth, due to the studies available, ST alone could not be analyzed. Thus, future research is needed in this area and should explore the effects of ST in comparison with AE to offer a more balanced perspective on PA guidelines during pregnancy. Additionally, other maternal, pregnancy, and birth outcome variables were not addressed in this review; thus, further research is needed.

5 | CONCLUSION

ST programs of more than 12 weeks, with 1–2 weekly sessions at moderate-to-vigorous intensity, significantly improve HRQoL and associated outcomes in PW. These findings highlight the importance of implementing well-designed ST programs into prenatal care to enhance maternal well-being. However, the limited evidence, particularly regarding ST in the first trimester, underscores the need for further research. Additionally, the heterogeneity in study outcomes and methodologies restricts definitive recommendations. Future studies should adopt standardized protocols to strengthen the evidence base and ensure consistent guidelines for PW.

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

PW should participate in a structured and supervised ST program administered by sport science professionals. Based on the results of our systematic review, an effective regimen could comprise ST sessions at least 1–2 days per week, each lasting 60 min. The session should be organized as a circuit-based training by time with functional exercises for both the upper and lower body (e.g., squats, lunges, deadlift, shoulder press, or row), with particular emphasis on the core and pelvic floor. According to the latest position paper by the European Board and College of Obstetrics and Gynaecology,[57] participants should maintain a moderate exercise intensity based on RPE (12–14 out of 20). Mobility exercises should be implemented during pregnancy, with increased focus in the last trimester to prepare the body for childbirth.

AUTHOR CONTRIBUTIONS

Paula Redondo-Delgado: Conceptualization, methodology, formal analysis, and writing—original draft preparation. Paula Blanco-Giménez: data curation and writing—review and editing. Susana López-Ortiz: methodology, software, validation, and formal analysis. Celia García-Chico: validation and formal analysis. Juan Vicente-Mampel: methodology and writing—original draft preparation. Sergio Maroto-Izquierdo: Conceptualization, methodology, validation, and writing—review and editing. All authors have read and agreed to the published version of the manuscript.

FUNDING INFORMATION

Research by PR-D was supported by the Ministry of Education of Spain (grant number 24CO1/001933). Research by CG-C is funded by Miguel de Cervantes European University, Department of Health Sciences, i+HeALTH Strategic Research Group. The publication of the work has been funded by the Catholic University of Valencia.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ORCID

Paula Redondo-Delgado https://orcid.org/0009-0007-5855-9762

REFERENCES

1. World Health Organization. *WHO guidelines on physical activity and sedentary behaviour [Internet]*. WHO; 2020. https://iris.who.int/bitstream/handle/10665/336656/9789240015128-eng.pdf?sequence=1
2. ACOG. Committee opinion No. 650: physical activity and exercise during pregnancy and the postpartum period. *Obstet Gynecol*. 2015;126:e135-e142.
3. Mottola MF, Davenport MH, Ruchat SM, et al. 2019 Canadian guideline for physical activity throughout pregnancy. *Br J Sports Med*. 2018;52:1339-1346.
4. Downs DS, Chasan-Taber L, Evenson KR, Leiferman J, Yeo S. Physical activity and pregnancy. *Res Q Exerc Sport*. 2012;83:485-502.
5. Fazzi C, Saunders DH, Linton K, Norman JE, Reynolds RM. Sedentary behaviours during pregnancy: a systematic review. *Int J Behav Nutr Phys Act*. 2017;14:32.
6. Salari N, Mohammadi A, Hemmati M, et al. The global prevalence of low back pain in pregnancy: a comprehensive systematic review and meta-analysis. *BMC Pregnancy Childbirth*. 2023;23:830.
7. Littleton HL, Breitkopf CR, Berenson AB. Correlates of anxiety symptoms during pregnancy and association with perinatal outcomes: a meta-analysis. *Am J Obstet Gynecol*. 2007;196:424-432.
8. Becker M, Weinberger T, Chandy A, Schmukler S. Depression during pregnancy and postpartum. *Curr Psychiatry Rep*. 2016;18:32.
9. Kominiarek MA, Peaceman AM. Gestational weight gain. *Am J Obstet Gynecol*. 2017;217:642-651.
10. Harrison AL, Taylor NF, Shields N, Frawley HC. Attitudes, barriers and enablers to physical activity in pregnant women: a systematic review. *J Physiother*. 2018;64:24-32.
11. Lagadec N, Steinecker M, Kapassi A, et al. Factors influencing the quality of life of pregnant women: a systematic review. *BMC Pregnancy Childbirth*. 2018;18:455.
12. Puspitasari N. Development of indicators to measure quality of life for pregnant women (QOL-PW). *J Prev Med Hyg*. 2023;64:E55-E66.
13. Wójcik M, Aniśko B, Siatkowski I. Quality of life in women with normal pregnancy. *Sci Rep*. 2024;14:12434.
14. Wu H, Sun W, Chen H, et al. Health-related quality of life in different trimesters during pregnancy. *Health Qual Life Outcomes*. 2021;19:182.
15. Karimi M, Brazier J. Health, health-related quality of life, and quality of life: what is the difference? *PharmacoEconomics*. 2016;34:645-649.
16. Bai G, Korfage IJ, de Groen EH, Jaddoe VWV, Mautner E, Raat H. Associations between nausea, vomiting, fatigue and health-related quality of life of women in early pregnancy: the generation R study. *PLoS One*. 2016;11(11):e0166133. doi:10.1371/journal.pone.0166133
17. Regan AK, Swathi PA, Nosek M, Gu NY. Measurement of health-related quality of life from conception to postpartum using the EQ-5D-5L among a National Sample of US pregnant and postpartum adults. *Appl Health Econ Health Policy*. 2023;21:523-532.
18. Streuling I, Beyerlein A, Rosenfeld E, Hofmann H, Schulz T, von Kries R. Physical activity and gestational weight gain: a meta-analysis of intervention trials. *BJOG*. 2011;118:278-284.
19. Davenport MH, Marchand AA, Mottola MF, et al. Exercise for the prevention and treatment of low back, pelvic girdle and lumbopelvic pain during pregnancy: a systematic review and meta-analysis. *Br J Sports Med*. 2019;53:90-98.
20. Barakat R, Pelaez M, Montejo R, Luaces M, Zakynthinaki M. Exercise during pregnancy improves maternal health perception: a randomized controlled trial. *Am J Obstet Gynecol*. 2011;204:402.e1-402.e7.
21. O'Connor PJ, Poudevigne MS, Johnson KE, Brito de Araujo J, Ward-Ritacco CL. Effects of resistance training on fatigue-related domains of quality of life and mood during pregnancy: a randomized trial in pregnant women with increased risk of Back pain. *Psychosom Med*. 2018;80:327-332.
22. Krzepota J, Sadowska D, Biernat E. Relationships between physical activity and quality of life in pregnant women in the second and third trimester. *Int J Environ Res Public Health*. 2018;15:2745.
23. Liu N, Gou W h, Wang J, et al. Effects of exercise on pregnant women's quality of life: a systematic review. *Eur J Obstet Gynecol Reprod Biol*. 2019;242:170-177. doi:10.1016/j.ejogrb.2019.03.009
24. Díaz-Burrueco JR, Cano-Ibáñez N, Martín-Peláez S, Khan KS, Amezcua-Prieto C. Effects on the maternal-fetal health outcomes of various physical activity types in healthy pregnant women. A systematic review and meta-analysis. *Eur J Obstet Gynecol Reprod Biol*. 2021;262:203-215. doi:10.1016/j.ejogrb.2021.05.030
25. Sánchez-Polán M, Nagpal TS, Zhang D, Silva-Jose C, Montejo R, Barakat R. The influence of physical activity during pregnancy on maternal pain and discomfort: a meta-analysis. *J Pers Med*. 2023;14:44.
26. Dipietro L, Evenson KR, Bloodgood B, et al. Benefits of physical activity during pregnancy and postpartum: an umbrella review. *Med Sci Sports Exerc*. 2019;51:1292-1302.

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

27. Higgins JPT, Thomas J, Chandler J, et al., eds. *Cochrane handbook for systematic reviews of interventions [Internet]. Version 6.5*. Cochrane; 2024 www.training.cochrane.org/handbook
28. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. doi:10.1136/bmj.n71
29. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898.
30. Smart NA, Waldron M, Ismail H, et al. Validation of a new tool for the assessment of study quality and reporting in exercise training studies. *Int J Evid Based Healthc*. 2015;13:9-18.
31. Acosta-Manzano P, Acosta FM, Coll-Risco I, et al. The influence of exercise, lifestyle behavior components, and physical fitness on maternal weight gain, postpartum weight retention, and excessive gestational weight gain. *Int J Sport Nutr Exerc Metab*. 2022;32:425-438.
32. Aparicio VA, Marín-Jiménez N, Flor-Alemany M, Acosta-Manzano P, Coll-Risco I, Baena-García L. Effects of a concurrent exercise training program on low back and sciatic pain and pain disability in late pregnancy. *Scand J Med Sci Sports*. 2023;33:1201-1210.
33. Petrov FK, Glantz A, Fagevik Olsen M. The efficacy of moderate-to-vigorous resistance exercise during pregnancy: a randomized controlled trial. *Acta Obstet Gynecol Scand*. 2015;94:35-42.
34. Haakstad LAH, Torset B, Bø K. What is the effect of regular group exercise on maternal psychological outcomes and common pregnancy complaints? An assessor blinded RCT. *Midwifery*. 2016;32:81-86.
35. Haakstad LAH, Bø K. Effect of regular exercise on prevention of excessive weight gain in pregnancy: a randomised controlled trial. *Eur J Contracept Reprod Health Care*. 2011;16(2):116-125. doi:10.3109/13625187.2011.560307
36. Haakstad L, Bø K. Effect of a regular exercise programme on pelvic girdle and low back pain in previously inactive pregnant women: a randomized controlled trial. *J Rehabil Med*. 2015;47:229-234.
37. Marín-Jiménez N, Baena-García L, Coll-Risco I, Flor-Alemany M, Castro-Piñero J, Aparicio VA. Influence of a concurrent exercise training program on health-related quality of life during advanced pregnancy: the GESTAFIT project. *Sports Health*. 2024; 16:518-526.
38. Ruiz JR, Perales M, Pelaez M, Lopez C, Lucia A, Barakat R. Supervised exercise–based intervention to prevent excessive gestational weight gain: a randomized controlled trial. *Mayo Clin Proc*. 2013;88:1388-1397.
39. Aparicio VA, Ocón O, Padilla-Vinuesa C, et al. Effects of supervised aerobic and strength training in overweight and grade I obese pregnant women on maternal and foetal health markers: the GESTAFIT randomized controlled trial. *BMC Pregnancy Childbirth*. 2016;29(16):290.
40. Slavin V, Gamble J, Creedy DK, Fenwick J, Pallant J. Measuring physical and mental health during pregnancy and postpartum in an Australian childbearing population—validation of the PROMIS global short form. *BMC Pregnancy Childbirth*. 2019;19:370.
41. Boutib A, Chergaoui S, Marfak A, Hilali A, Youlyouz-Marfak I. Quality of life during pregnancy from 2011 to 2021: systematic review. *Int J Women's Health*. 2022;14:975-1005.
42. Shakeel N, Richardsen KR, Martinsen EW, Eberhard-Gran M, Slinning K, Jenum AK. Physical activity in pregnancy and postpartum depressive symptoms in a multiethnic cohort. *J Affect Disord*. 2018;236:93-100.
43. Padmapriya N, Bernard JY, Liang S, et al. Association of physical activity and sedentary behavior with depression and anxiety symptoms during pregnancy in a multiethnic cohort of Asian women. *Arch Womens Ment Health*. 2016;19:1119-1128.
44. Coll C de VN, Domingues MR, Stein A, et al. Efficacy of regular exercise during pregnancy on the prevention of postpartum depression. *JAMA Netw Open*. 2019;2:e186861.
45. Duchette C, Perera M, Arnett S, White E, Belcher E, Tinius R. Benefits of resistance training during pregnancy for maternal and fetal health: a brief overview. *Int J Women's Health*. 2024;16:1137-1147.
46. Haakstad L, Torset B, Bo K. Beyond the focus of excessive weight gain during pregnancy. *Med Sci Sports Exerc*. 2014;46:722-723.
47. Prevett C, Kimber ML, Forner L, de Vivo M, Davenport MH. Impact of heavy resistance training on pregnancy and postpartum health outcomes. *Int Urogynecol J*. 2023;34:405-411.
48. Kuffel EE, Pfeiffer KA, Holzman CB, Pivarnik JM. Associations between resistance training during first trimester on pregnancy and birth outcomes. *Med Sci Sports Exerc*. 2011;43:157.
49. Kokic S, Kokic I, Ivanisevic M, et al. Combination of a structured aerobic and resistance exercise improves glycaemic control in pregnant women diagnosed with gestational diabetes mellitus. A randomised controlled trial. *Women Birth*. 2018;31:e232-e238.
50. Yaping X, Huifen Z, Chunhong L, Fengfeng H, Huibin H, Meijing Z. A meta-analysis of the effects of resistance training on blood sugar and pregnancy outcomes. *Midwifery*. 2020;91:102839.
51. Corso M, Miarka B, Figueiredo T, Bragazzi N, Carvalho D, Dias I. Effects of aerobic, strength, and combined training during pregnancy in the blood pressure: a systematic review and meta-analysis. *Front Physiol*. 2022;13:916724.
52. Murphy SE, Johnston CA, Strom C, et al. Influence of exercise type on maternal blood pressure adaptation throughout pregnancy. *AJOG Global Rep*. 2022;2:100023.
53. Ward-Ritacco C, Poudevigne MS, Oconnor PJ. Muscle strengthening exercises during pregnancy are associated with increased energy and reduced fatigue. *J Psychosom Obstet Gynecol*. 2016;37:68-72.
54. Oksanen P, Tikanmäki M, Tulppo MP, Niemelä M, Korpelainen R, Kajantie E. Physical activity, fitness, and cardiac autonomic function among adults born postterm. *Am J Epidemiol*. 2025;194:766-778.
55. Yenişehir S, Çıtak Karakaya İ, Sivaslıoğlu AA, Özen Oruk D, Karakaya MG. Reliability and validity of five times sit to stand test in pregnancy-related pelvic girdle pain. *Musculoskelet Sci Pract*. 2020;48:102157.
56. O'Connor PJ, Poudevigne MS, Cress ME, Motl RW, Clapp JF. Safety and efficacy of supervised strength training adopted in pregnancy. *J Phys Act Health*. 2011;8:309-320.
57. Harmsworth M, Savona-Ventura C, Mahmood T. High-intensity exercise during pregnancy—a position paper by the European board and College of Obstetrics and Gynaecology (EBCOG). *Eur J Obstet Gynecol Reprod Biol*. 2023;285:56-58.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Redondo-Delgado P, Blanco-Giménez P, López-Ortiz S, García-Chico C, Vicente-Mampel J, Maroto-Izquierdo S. Effects of strength training on quality of life in pregnant women: A systematic review. *Acta Obstet Gynecol Scand*. 2025;104:1231-1243. doi:10.1111/aogs.15122

16000412, 2025, 7, Downloaded from https://obgyn.onlinelibrary.wiley.com/doi/10.1111/aogs.15122 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [08/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License